Numericals

(Chapter 12)

Q1

· Data:-

$y(x,t) = 3.5 \sin\left\{\frac{\pi}{3}x - 66t\right\}$ cm

Amplitude = x_o = ?
wavelength = λ = ?

Sol:-

$\because y = A \sin(kx - \omega t) \rightarrow$ (i)

and

$y = 3.5 \sin\left(\frac{\pi}{3}x - 66t\right) \rightarrow$ (ii)

comparing eq (i) and (ii)

$\boxed{A = 3.5 = x_o}$

and $k = \frac{\pi}{3}$

wave number :- i) number of wave per unit distance.
ii) Reciprocal of wavelength

wave no. = $k = \frac{2\pi}{\lambda}$

since, $k = 2\pi/\lambda$

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Note: If you find any mistake in the pdf report it to NCA

$$\therefore \frac{2\pi}{7} = \frac{\pi}{3}$$

$$\boxed{\lambda = 6cm}$$

Q4

Data:-

$\Delta P = 100 kPa = 100 \times 1000 = 10^5 Pa$

$\frac{\Delta V}{V} \% = 5 \times 10^{-3} \% = 5 \times 10^{-5}$

Bulk modulus $= B = ?$

speed of sound in water $= v = ?$

Sol:-

$$\therefore B = \frac{\Delta P}{\frac{\Delta V}{V}} = \frac{10^5}{5 \times 10^{-5}} = 2 \times 10^9$$

$$B = 2000 \times 10^6$$

$$\boxed{B = 2000 MPa}$$

$$\therefore v = \sqrt{\frac{B}{\rho}}$$

$$v = \sqrt{\frac{2 \times 10^9}{1000}}$$

$$\boxed{v = 1414.21}$$

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Q5

$l = 10m$

weight of string $= w = 0.25N$

weight hang $= w' = 1KN = 1000N$

$t = ?$

Sol:

$$\therefore t = \frac{S}{V}$$

1st find 'v'

$$V = \sqrt{\frac{T}{\mu}}$$

$T = w' = 1000N$

$$\mu = \frac{m}{L} = \frac{w/g}{L} = \frac{w}{gl} = \frac{0.25}{9.8 \times 10}$$

$\mu = 2.55 \times 10^{-3}$

$$V = \sqrt{\frac{1000}{2.55 \times 10^{-3}}}$$

$V = 626.2 \, m/s$

$$\therefore t = \frac{S}{V} = \frac{10}{626.2}$$

$t = 0.0159$ sec

(or)

$t = 15.9$ msec

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Q6

Data:-

Amplitude = $A = 5cm$

Resultant Amplitude = $Y = 6.69\ cm$

phase difference = $\phi = ?$

Sol:-

$$y = 2A_o \cos\frac{\phi}{2}$$

$$\cos^{-1}\left(\frac{y}{2A_o}\right) = \frac{\phi}{2}$$

$$\cos^{-1}\left(\frac{6.69}{2(5)}\right) =$$

$$48 \times 2 = \phi$$

$$\phi = 96^\circ$$

Q7

Data:-

origional frequency = f_1

change in frequency = $f' = \frac{100-4}{100} = 0.96f_1$

% decrease in tension = $\frac{\Delta T}{T} = ?$

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Sol:-

since $f = \frac{1}{2l}\sqrt{\frac{T}{\mu}}$

$\therefore f \propto \sqrt{T}$

$f_1 = k\sqrt{T}$

Similarly

$f' = k\sqrt{T'}$

divide eq (ii) by (i)

$\Rightarrow \frac{f'}{f_1} = \sqrt{\frac{T'}{T}}$

$\Rightarrow \frac{0.96\cancel{f_1}}{\cancel{f_1}} = \sqrt{\frac{T'}{T}}$

Taking sq o.b.s

$\Rightarrow (0.96)^2 = \left(\cancel{\sqrt{\frac{T'}{T}}}\right)^{\cancel{2}}$

$\Rightarrow \boxed{T(0.96)^2 = T'}$

$\therefore \frac{\Delta T}{T}\% = \frac{T - T'}{T} \times 100$

$\therefore \frac{\Delta T}{T}\% = \frac{T - T(0.96)^2}{T} \times 100$

$$\Rightarrow \frac{\Delta T}{T}\% = \frac{\cancel{T}(1-0.96^2)}{\cancel{T}} \times 100$$

$$\Rightarrow \frac{\Delta T}{T}\% = 7.84\%$$

$\Rightarrow$ % decrease in tension = 7.84 %

Q8

Data:-

$L = 2m$

$t = 0.05\,sec$

Sol:-

$$\therefore f_1 = \frac{\cancel{L}/t}{2\cancel{L}}$$

$$\Rightarrow f_1 = \frac{1}{2t}$$

$$\Rightarrow f_1 = \frac{1}{2(0.05)}$$

$$\Rightarrow f_1 = 10\ Hz$$

Q10

Data:-

$f' = 219Hz$ (moving towards observer)

Q $f'' = 184Hz$ (moving away from observer)

$V = 340m/s$

$V_s = ?$

$f = ?$

Sol:- train moving towards the listener:

(formula) $f' = \frac{V}{V-V_s} \cdot f_s \rightarrow (i)$

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train moving away from the listener :

$$\Rightarrow \quad f'' = \frac{v}{v + V_s} \cdot f_s \rightarrow (ii)$$

divide eq (ii) by (i)

$$\Rightarrow \quad \frac{f'}{f''} = \frac{v + v_s}{v - v_s}$$

$$\Rightarrow \quad \frac{219}{184} = \frac{340 + V_s}{340 - V_s}$$

$$219(340 - V_s) = 184(340 + V_s)$$

$$74460 - 219V_s = 62560 + 184V_s$$

$$74460 - 62560 = 184V_s + 219V_s$$

$$11900 = 403\,V_s$$

$$V_s = \frac{11900}{403}$$

$$\boxed{V_s = 29.52 \text{ m/s}}$$

put in eq (i)

$$219 = \frac{340}{340 - 29.52} \cdot f$$

$$219 \times 310.48 = 340f$$

$$f = \frac{219 \times 310.48}{340}$$

$$\boxed{f = 200\,\text{Hz}}$$